

# Sensorless Control for an Open-Phase Five-Phase PMSM by Using Improved SOGIs and an Adaptive Neutral-Voltage Compensation

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The occurrence of open-phase faults in five-phase permanent magnet synchronous motors (5Ph-PMSMs) disrupts the current balance among the remaining healthy phases, resulting in the introduction of second and fourth harmonics into the position estimation. To address this challenge, this paper proposes an innovative sensorless control approach specifically for open-phase 5Ph-PMSM systems. This methodology integrates improved second-order generalized integrators (ISOGIs) to effectively mitigate undesirable harmonics and extract the positive sequence component from the distorted back electromotive force (back-EMF). Additionally, an adaptive neutral-voltage compensator is implemented to alleviate the asymmetry present in the back-EMF, thereby further enhancing the precision of the sensorless control. The effectiveness of the proposed methodology is corroborated through rigorous experimental results.

**Keywords:** five-phase motor, open-phase fault, sensorless control, second-order generalized integrator

## 1. Introduction

Permanent Magnet Synchronous Motors (PMSMs) occupy a prominent position in many industry sectors. In comparison to a three-phase PMSM, five-phase PMSMs (5Ph-PMSMs) exhibit significant advantages in their higher power density, enhanced control flexibility, and reduced torque ripple<sup>(2)(3)</sup>. Due to these merits, 5Ph-PMSMs are increasingly favored in applications such as aerospace and electric vehicles. Notably, in the event of open-phase faults, the 5Ph-PMSMs possess the capability to directly enter fault-tolerant control by merely reconfiguring the current in the remaining healthy phases, thereby eliminating the need for additional hardware<sup>(3)</sup>. Furthermore, position sensors are also prone to failure, and in order to mitigate potential failures in field-oriented control (FOC), the sensorless control technique has become a core algorithm in engineering practices<sup>(4)</sup>. To further bolster the reliability of 5Ph-PMSMs, it is imperative to conduct research on sensorless control under open-phase fault conditions.

Currently, quite a few studies have been conducted concerning the sensorless control of multi-phase motors under open-phase fault conditions. In this regard, reference (5) introduced a sensorless control technology grounded in a robust wide-speed sliding mode observer (SMO), which is attempted in a novel five-phase fault-tolerant motor. In the event of an open-phase fault, this method can also seamlessly

transition from traditional space vector pulse width modulation (SVPWM) to a fault-tolerant SVPWM. Additionally, reference (6) investigated the effects of open-phase faults on the high-frequency injection-based sensorless control for a dual three-phase motor. To simplify the analysis process, a high-frequency fault analysis method is presented in (6), along with an equivalent coordination method to extend the results to arbitrary single open-phase faults. Building on this, a new rotor position estimation method based on a modified adaptive filter is introduced, which is robust to open-phase faults without requiring any fault diagnosis. Moreover, reference (7) introduced a novel sensorless control method that utilizes a robust observer, a nonorthogonal phase-locked loop (PLL), and variable phase delay compensation. This approach ensures medium- and high-speed sensorless control performance for fault-tolerant motors, even in the presence of phase open-circuit and short-circuit fault conditions.

This paper attempts to improve the sensorless control performance of open-phase 5Ph PMSMs from the following two aspects. Firstly, low-pass filters (LPFs) in the conventional sliding mode sensorless control are replaced by second-order generalized integrators (SOGIs) to extract the fundamental frequency components from the rippled back electromotive force (EMF) estimate. Considering the negative sequence components caused by an open-phase fault, an enhanced extraction method is constructed by using improved SOGIs (ISOGIs) for calculating the positive sequence as well as a PLL for calculating rotor position. Secondly, an adaptive neutral-voltage compensation method, by making use of the amplitude difference of the ISOGIs outputs, is adopted to equilibrate  $\alpha$ - $\beta$  frame back-EMF estimates, further reducing the periodical position estimation error in the sensorless control.

This paper is based on Reference (1), which published in ICEMS 2024 (c) 2024 IEEJ.

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## 2. Five-Phase PMSM Model under Fault

### 2.1 Decoupling Model under Open-Phase Fault

Similar to three-phase PMSMs, 5Ph-PMSM model can also be decomposed into the synchronous rotating reference frame from the angle of FOC. However, when an open-phase fault occurs in a 5Ph-PMSM, the normal mode decoupling matrix is no longer applicable due to lack of excitation in the opened phase. Without losing generality, this paper takes an A-phase open fault as an example. Based on the principle that the magneto-motive force should be kept unchanged before and after the fault, the decoupling matrix after an A-phase open fault needs to be corrected. According to the analysis process in Reference (3), the reduced-order matrix after the fault can be obtained as follows.

The reduced-order Clarke matrix in post-fault is below

$$\mathbf{T}_1 = \frac{2}{5} \begin{bmatrix} \cos \delta - 1 & \cos 2\delta - 1 & \cos 3\delta - 1 & \cos 4\delta - 1 \\ \sin \delta & \sin 2\delta & \sin 3\delta & \sin 4\delta \\ \sin 3\delta & \sin 6\delta & \sin 9\delta & \sin 12\delta \\ 1 & 1 & 1 & 1 \end{bmatrix} \dots \dots \dots (1)$$

with  $\delta = 2\pi/5$ .

The reduced-order Park matrix in post-fault is below

$$\mathbf{T}_2 = \begin{bmatrix} \cos \theta_e & \sin \theta_e & 0 & 0 \\ -\sin \theta_e & \cos \theta_e & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \dots \dots \dots (2)$$

Based on the reduced-order transformation matrices (1) and (2), the single-phase opened 5Ph-PMSM can be modelled in fault tolerant d-q frame as follows.

$$\begin{bmatrix} u_{d1} \\ u_{q1} \\ u_{q3} \end{bmatrix} = R_s \begin{bmatrix} i_{d1} \\ i_{q1} \\ i_{q3} \end{bmatrix} + \begin{bmatrix} L_{d1} & 0 & 0 \\ 0 & L_{q1} & 0 \\ 0 & 0 & L_{q3} \end{bmatrix} \frac{d}{dt} \begin{bmatrix} i_{d1} \\ i_{q1} \\ i_{q3} \end{bmatrix} - \omega_e \begin{bmatrix} 0 & L_{q1} & 0 \\ -L_{d1} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} i_{d1} \\ i_{q1} \\ i_{q3} \end{bmatrix} + \omega_e \begin{bmatrix} 0 \\ \psi_{m1} \\ 3\psi_{m3} \cos 3\theta_e \end{bmatrix} \dots \dots \dots (3)$$

where,  $u_{d1}$ ,  $u_{q1}$ ,  $u_{q3}$  and  $i_{d1}$ ,  $i_{q1}$ ,  $i_{q3}$  are, respectively, the voltages and currents under the fault tolerant d-q frame (The subscripts d1 and q1 signify components projected onto the fundamental synchronous reference frame, while the subscript q3 denotes the component projected onto the third-harmonic synchronous reference frame);  $L_{d1}$ ,  $L_{q1}$  and  $L_{q3}$  are the inductance projection of the d1, q1, and q3 axes, respectively;  $\omega_e$  is the electrical angular velocity of the motor rotor;  $\psi_{m1}$  and  $\psi_{m3}$  are the amplitudes of the first and third order flux linkages.

**2.2 Neutral Voltage Deviation and Compensation Method** In the event of open-phase fault, the remaining healthy phase voltage of the motor can be represented by the switching status combination as below

$$\begin{bmatrix} u_{BN} \\ u_{CN} \\ u_{DN} \\ u_{EN} \end{bmatrix} = U_{dc} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} S_B \\ S_C \\ S_D \\ S_E \end{bmatrix} - \left( \frac{U_{dc}}{2} + u_{NO} \right) \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} \dots \dots \dots (4)$$

where  $U_{dc}$  is the DC bus voltage;  $u_{NO}$  is the potential difference between the neutral point of the motor and the midpoint of the DC bus;  $S_X$  is the switching status, with  $X = B, C, D, E$ , and  $S_X = 1$  indicating the upper IGBT of X-Bridge is turned on, and the lower IGBT complementary to it.

On the premise that the motor structure is not damaged, one can have the following zero-sum constraint for the five phase voltages.

$$u_{AN} + u_{BN} + u_{CN} + u_{DN} + u_{EN} = 0 \dots \dots \dots (5)$$

Substituting (4) and (5), it yields

$$u_{NO} = \frac{U_{dc}}{4} (S_B + S_C + S_D + S_E) - \frac{U_{dc}}{2} + \frac{1}{4} u_{AN} \dots \dots \dots (6)$$

From (6),  $u_{NO}$  can change with  $u_{AN}$  under the open-phase fault, and this implies the phase voltage is not only dictated to its respective modulation voltage but also  $u_{NO}$ . According to (1), (4) and (6), the modulation voltage under the fault tolerant  $\alpha$ - $\beta$  frame can be represented by

$$\begin{bmatrix} u_{\alpha 1} \\ u_{\beta 1} \\ u_{\beta 3} \end{bmatrix} = U_{dc} \mathbf{T}_3 \begin{bmatrix} S_B \\ S_C \\ S_D \\ S_E \end{bmatrix} + \begin{bmatrix} 0.5u_{AN} \\ 0 \\ 0 \end{bmatrix} \dots \dots \dots (7)$$

with

$$\mathbf{T}_3 = \frac{2}{5} \begin{bmatrix} \cos \delta + \frac{1}{4} & \cos \delta + \frac{1}{4} & \cos \delta + \frac{1}{4} & \cos \delta + \frac{1}{4} \\ \sin \delta & \sin 2\delta & \sin 3\delta & \sin 4\delta \\ \sin 3\delta & \sin 6\delta & \sin 9\delta & \sin 12\delta \end{bmatrix} \dots \dots \dots (8)$$

To mitigate the influence of  $u_{AN}$  on  $u_{\alpha 1}$ , the following compensation method can be adopted

$$\begin{cases} u_{comp} = 0.5u_{AN} \\ v_{\alpha 1} = u_{\alpha 1} - u_{comp} \\ v_{\beta 1} = u_{\beta 1} \\ v_{\beta 3} = u_{\beta 3} \end{cases} \dots \dots \dots (9)$$

## 3. Proposed Sensorless Control Method

**3.1 Sliding Mode Observer** With the aforementioned reduced-order Clarke and Park transformations, the fundamental space is supposed to be maximally decoupled from the third harmonic space. On the other hand, for the studied surface-mounted PMSM, one has  $L_{d1} = L_{q1} = L$ . Thus, by taking into account the fundamental space model, a current state equation can be built as below

$$\begin{cases} \frac{di_{\alpha}}{dt} = \frac{1}{L} (u_{\alpha} - Ri_{\alpha} - e_{\alpha}) \\ \frac{di_{\beta}}{dt} = \frac{1}{L} (u_{\beta} - Ri_{\beta} - e_{\beta}) \end{cases} \dots \dots \dots (10)$$

Meanwhile, as per previous articles, the sliding mode observer can usually be designed as follows

$$\begin{cases} \frac{d\hat{i}_{\alpha}}{dt} = \frac{1}{L} (u_{\alpha} - R\hat{i}_{\alpha} - z_{\alpha}) \\ \frac{d\hat{i}_{\beta}}{dt} = \frac{1}{L} (u_{\beta} - R\hat{i}_{\beta} - z_{\beta}) \end{cases} \dots \dots \dots (11)$$

where  $\hat{i}_\alpha$  and  $\hat{i}_\beta$  are the  $\alpha$ - $\beta$  frame motor currents;  $z_\alpha$  and  $z_\beta$  are the control laws which can be designed as the following

$$\begin{cases} z_\alpha = h \cdot \text{sign}(\tilde{i}_\alpha) \\ z_\beta = h \cdot \text{sign}(\tilde{i}_\beta) \end{cases}, \quad h > \max(|e_\alpha|, |e_\beta|) \dots \dots \dots (12)$$

with  $\tilde{i}_\alpha = \hat{i}_\alpha - i_\alpha$ ,  $\tilde{i}_\beta = \hat{i}_\beta - i_\beta$ ; and  $h$  stands for the sliding mode gain.

**3.2 ISOGI-FLL with PLL for Rotor Position Extraction** In the open-phase fault mode, the healthy phase is asymmetrically powered, and therefore the observed back EMF may carry many unwanted harmonics, including the negative sequence. This is easy to understand when recalling an unevenly loaded three-phase drive, where both positive and negative sequences exist simultaneously. In this regard, this paper proposes a positive sequence extraction method to aid in the back EMF estimation.

$$e_{abcde}^+ = \begin{bmatrix} 1 & \alpha & \alpha^2 & \alpha^3 & \alpha^4 \\ \alpha^4 & 1 & \alpha & \alpha^2 & \alpha^3 \\ \alpha^3 & \alpha^4 & 1 & \alpha & \alpha^2 \\ \alpha^2 & \alpha^3 & \alpha^4 & 1 & \alpha \\ \alpha & \alpha^2 & \alpha^3 & \alpha^4 & 1 \end{bmatrix} e_{abcde} \dots \dots \dots (13)$$

where  $\alpha = e^{j2\pi/5}$ . With the Clarke transformation, the positive sequence component in a stationary reference frame can be expressed as

$$\begin{bmatrix} e_\alpha^+ \\ e_\beta^+ \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 1 & -q \\ q & 1 \end{bmatrix} \begin{bmatrix} e_\alpha \\ e_\beta \end{bmatrix} \dots \dots \dots (14)$$

where  $q = e^{-j\pi/2}$ , and it represents a  $90^\circ$  phase shift from the original signal. From (14), before extracting the positive sequence, one needs to acquire the signals  $e_\alpha$  and  $e_\beta$  as well as their respective orthogonal signals in advance.

SOGI has been widely used in grid-tied inverter, and its block diagram is shown in Fig. 1. Let's take the acquisition of  $e_\alpha$  and  $qe_\alpha$  as an example for analysis. The input signal  $z_\alpha$  is sinusoidal, and it outputs two mutually orthogonal output signals  $\hat{e}_\alpha$  and  $q\hat{e}_\alpha$ .  $k_s$  is the gain of the forward path, which determines the bandwidth of the SOGI, while  $\omega_0$  is the resonant frequency of the SOGI. A smaller  $k_s$  results in a narrower bandwidth, which improves the frequency selectivity characteristics of the filter for input signals, but correspondingly degrades dynamic performance. Therefore, the selection of the  $k_s$  value requires a trade-off consideration.

The transfer function for SOGI is represented by the following two forms

$$D(s) = \frac{\hat{e}_\alpha(s)}{z_\alpha(s)} = \frac{k_s \omega_0 s}{s^2 + k_s \omega_0 s + \omega_0^2} \dots \dots \dots (15)$$

$$Q(s) = \frac{q\hat{e}_\alpha(s)}{z_\alpha(s)} = \frac{k_s \omega_0^2}{s^2 + k_s \omega_0 s + \omega_0^2} \dots \dots \dots (16)$$

Fig. 2 shows the amplitude-frequency and phase-frequency characteristics of  $D(s)$  and  $Q(s)$  when  $\omega_0 = 100$  rad/s. From the figure,  $D(s)$  functions as a band-pass filter, allowing the output signal  $\hat{e}_\alpha$  to track the input signal  $z_\alpha$  at the frequency of  $\omega_0$  without amplitude attenuation or phase delay.  $Q(s)$  is equivalent to a low-pass filter with its cutoff frequency being  $\omega_0$ , and  $Q(s)$  always lags  $D(s)$  by  $90^\circ$ , meaning that the output signals  $\hat{e}_\alpha$  and  $q\hat{e}_\alpha$  can be orthogonal to each other.

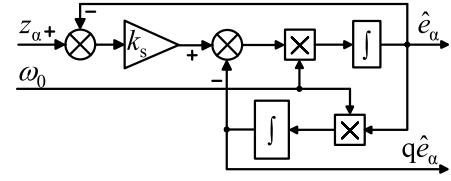


Fig. 1. Block diagram of SOGI

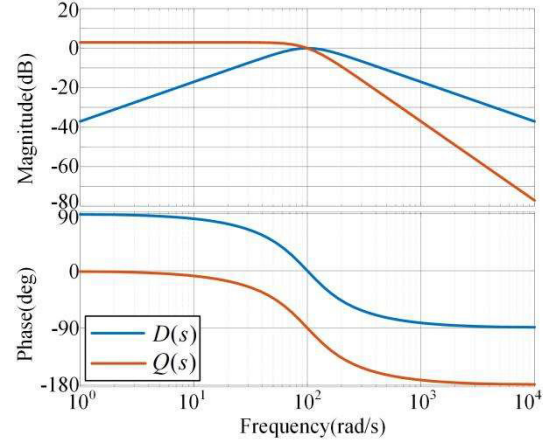


Fig. 2. Block diagram of  $D(s)$  and  $Q(s)$

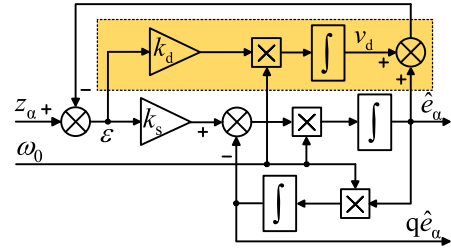


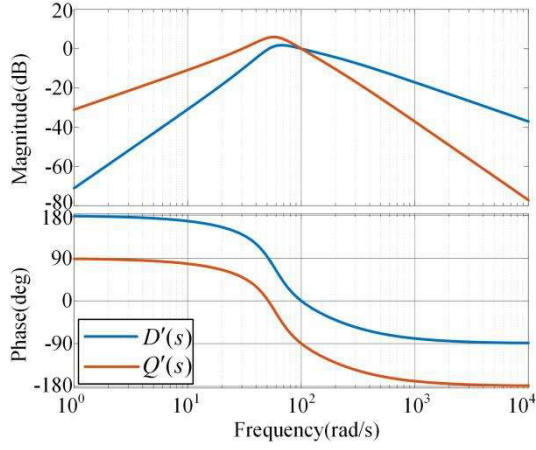
Fig. 3. Block diagram of ISOGI

However, when the input signal contains a direct current (DC) component, the low-pass filtering characteristic of  $Q(s)$  causes the quadrature signal output  $q\hat{e}_\alpha$  by the traditional SOGI to include the DC offset. In this case, the output signals  $\hat{e}_\alpha$  and  $q\hat{e}_\alpha$  no longer satisfy the condition of equal amplitude and orthogonality. Employing this structure for extracting the positive sequence component of back electromotive force (EMF) can lead to significant issues.

To address the issue of DC offset in traditional SOGI, this paper employs an improved SOGI (ISOGI) structure, the block diagram of which is shown in Figure 3. The ISOGI introduces an integral branch at the error signal  $\varepsilon$ , where the integral of the error signal  $v_d$  is summed with the output signal  $\hat{e}_\alpha$  and fed back to the input. The resulting error signal  $\varepsilon$ , obtained by subtracting the feedback from the input signal  $z_\alpha$ , no longer contains the DC component, thereby effectively mitigating the impact of the DC component on subsequent stages.

The transfer function for ISOGI is represented by the following two forms

$$D'(s) = \frac{\hat{e}_\alpha(s)}{z_\alpha(s)} = \frac{k_s \omega_0 s^2}{s^3 + (k_s + k_d) \omega_0 s^2 + \omega_0^2 s + k_d \omega_0^3} \dots \dots \dots (17)$$


 Fig. 4. Block diagram of  $D'(s)$  and  $Q'(s)$ 

$$Q'(s) = \frac{q\hat{e}_\alpha(s)}{z_\alpha(s)} = \frac{k_s \omega_0^2 s}{s^3 + (k_s + k_d)\omega_0 s^2 + \omega_0^2 s + k_d \omega_0^3} \dots (18)$$

Fig. 4 shows the amplitude-frequency and phase-frequency characteristics of  $D'(s)$  and  $Q'(s)$  when  $\omega_0 = 100$  rad/s. Compared to the traditional SOGI, the  $D'(s)$  of the ISOGI exhibits a steeper slope in the low-frequency range, indicating better suppression of DC components and low-frequency disturbances, which directly corresponds to improved band-pass filtering performance. Furthermore, the  $Q'(s)$  of the ISOGI transitions from a low-pass to a band-pass characteristic, thereby gaining the capability to suppress DC components to some extent. When the parameter  $k_d$  increases, the DC component suppression capability is enhanced. However, when  $k_d$  is set to a large value, it may lead to excessive oscillation in the output, thereby reducing system stability. Consequently, the selection of parameter  $k_d$  requires a balanced trade-off.

In AC speed regulation systems, the frequency of the fundamental back EMF varies with rotational speed. To ensure that the SOGI can accurately extract the fundamental back EMF over a wide speed range, a Frequency Locked Loop (FLL) must be incorporated into the SOGI structure. After adding the FLL, the angular velocity input to the ISOGI module is no longer  $\omega_0$ , but  $\omega_{fl}$ .

The response time of traditional FLL frequency estimation is affected by the amplitude of the input back EMF and the observed frequency. To address this, this paper adopts a per-unit normalized FLL with amplitude-frequency self-adaptation, which effectively eliminates the influence of back EMF amplitude and observed frequency on the FLL's response performance. The block diagram of the per-unit normalized FLL is shown in Fig. 5.

Assuming the fundamental frequency of the input signal is  $\omega$ , when  $\omega < \omega_{fl}$ ,  $\varepsilon_f > 0$ ; when  $\omega > \omega_{fl}$ ,  $\varepsilon_f < 0$ . The error signal is processed by negative gain and per-unit normalization before being fed into the integrator, thereby adjusting the  $\omega_{fl}$ . When  $\varepsilon_f = 0$ ,  $\omega = \omega_{fl}$ . This means that the FLL locks onto the fundamental frequency of the input signal.

The previous section uses the acquisition of  $e_\alpha$  and  $q e_\alpha$  as an example for analysis, and the principle for acquiring  $e_\beta$  and  $q e_\beta$  is the same as that. The block diagram of the positive-

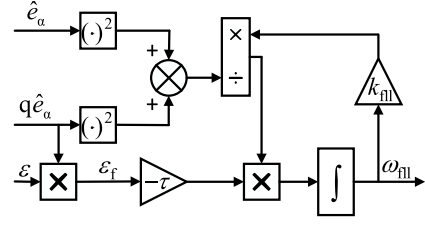


Fig. 5. Block diagram of per-unit normalized FLL

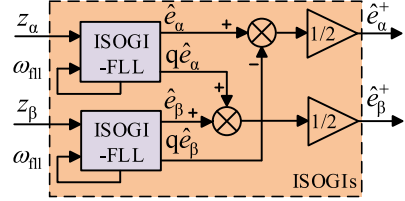


Fig. 6. Block diagram of the enhanced position estimation method by using ISOGI-FLL and PLL

sequence component extraction method is shown in Fig. 6.

By connecting the back EMF output of the SMO to the ISOGIs and applying (14), the positive sequence component of the back EMF can be extracted. Then, through PLL, the estimated position angle  $\hat{\theta}_e$  and the estimated angular velocity  $\hat{\omega}_e$  are calculated.

**3.3 Adaptive Neutral-Voltage Compensation** As mentioned earlier, in the context of an open-phase fault, the modulation voltage  $u_{\alpha 1}$  also needs fixing, where the compensation  $u_{comp}$  is related to  $u_{AN}$ . Considering most motor drives do not have any phase voltage sensor installed,  $u_{comp}$  needs to be calculated in other ways.

Because one has  $i_A \equiv 0$  in the absence of phase-A excitation,  $u_{AN}$  can be deemed equal to the back EMF of phase-A. On the other hand, the open-phase fault does not affect the symmetry of the stator winding distribution; therefore  $u_{AN}$  can be replaced by  $e_A$ , and by further substituting  $e_A$  into the symmetrical Clarke matrix, one has

$$e_{\alpha 1} = \frac{2}{5}(e_A + \cos \delta e_B + \cos 2\delta e_C + \cos 3\delta e_D + \cos 4\delta e_E) \dots (19)$$

In addition, the five back EMFs fulfil the following constraints

$$e_A + e_B + e_C + e_D + e_E = 0 \dots (20)$$

Substituting (19) into (20), it yields

$$e_A = e_{\alpha 1} = u_{AN} \dots (21)$$

From (21),  $\hat{e}_\alpha$  can be approximated as while performing neutral voltage compensation. However, this open-loop compensation depends on the accuracy of  $\hat{e}_\alpha$ , and to further reduce the compensation error, this paper introduces an adaptive compensation method below.

Fig. 7 shows the block diagram of the adaptive voltage compensation network proposed in this paper. Considering the steady state condition where  $\hat{e}_\alpha$  and  $\hat{e}_\beta$  are equal in magnitude and orthogonal to each other, it yields  $\hat{e}_\alpha + q\hat{e}_\beta = 0$ . At the occurrence of over/under compensation,  $\hat{e}_\alpha + q\hat{e}_\beta$  is featured by an amplitude error. By scaling up this error with

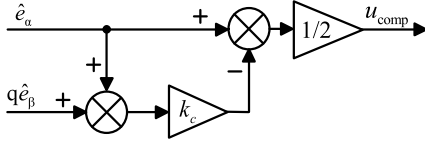


Fig. 7. Block diagram of the adaptive voltage compensation network

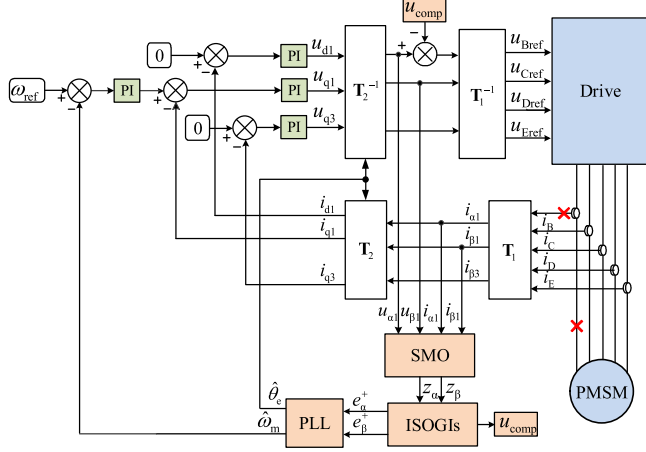


Fig. 8. Block diagram of the five-phase PMSM fault-tolerant control system under single-phase open-circuit fault

the gain  $k_c$  and then feeding it back to  $u_{comp}$ , the compensation error can gradually be eliminated. Considering the need to both ensure rapid response and avoid excessive regulation leading to oscillation, the  $k_c$  should be set to an appropriate value 0.05.

Based on the above content, the various modules of the motor control system have been introduced. Fig. 8 shows the structural block diagram of the five-phase PMSM fault-tolerant control system under single-phase open-circuit fault. The voltage and sampled current are input into the SMO module to estimate the back EMF. The ISOGIs module extracts the positive-sequence component from the back EMF and calculates the compensation voltage. The positive-sequence component of the back EMF is then fed into the PLL to estimate the rotor position angle and angular velocity. Additionally, the SPWM method is employed for modulation.

#### 4. Experiment Results

The proposed sensorless control method is validated on a laboratory-made 5Ph-PMSM rig, as shown in Fig. 9. The motor parameters of the 5Ph-PMSM are listed in Table 1. A three-phase rectifier bridge and power resistor are connected to load the three-phase generator, and an incremental encoder is installed to evaluate the proposed sensorless control performance.

Fig. 10 shows the back EMF estimate under two types of filters at 300 rpm, along with the waveforms of the rotor speed and position estimation errors. The LPF's cutoff frequency is set at 400 Hz, whilst the  $k_s$  of the SOGI is set to 2 and the  $k_d$  of the SOGI is set to 0.12. The results depicted in Fig. 10(b), which were obtained under SMO with SOGIs, demonstrate smoother outputs compared to those in Fig. 10(a), which utilized SMO with an LPF. Consequently,

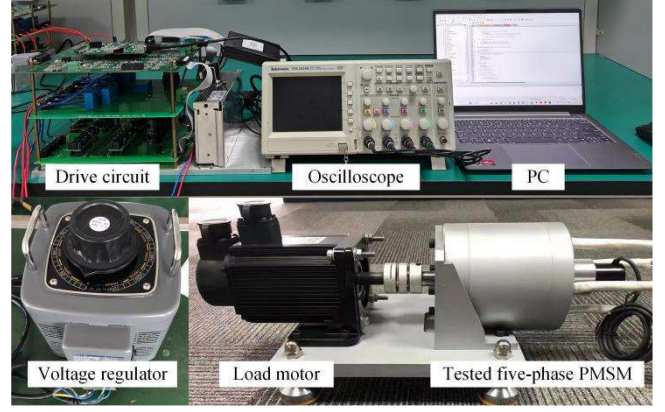


Fig. 9. 5Ph-PMSM experimental platform

Table 1. Motor Parameters

| Parameter            | Value        | Parameter     | Value |
|----------------------|--------------|---------------|-------|
| Number of pole-pairs | 2            | Rated voltage | 100V  |
| Stator resistance    | 0.4 $\Omega$ | Rated current | 10A   |
| d-q axis inductance  | 2.03mH       | Rated power   | 1kW   |

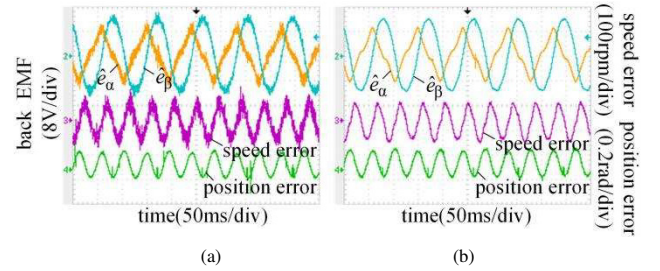


Fig. 10. Waveforms of the back EMF estimates, speed and position estimation error under (a) SMO combined with LPF; (b) SMO combined with ISOGIs

the fluctuation in speed estimation error through the PLL is notably reduced. This indicates that, when extracting the fundamental back EMF from high-frequency switching signals, the SOGIs demonstrates better performance over the LPF. Furthermore, Fig. 10 also reveals that when the motor operates under an open-phase fault condition, significant distortion appears in the back EMF estimate, accompanied by discernible 2nd harmonic in the rotor speed and position estimation errors.

Fig. 11 illustrates the extraction process of the positive sequence component from the back EMF estimate as well as the rotor position estimation behavior. From Fig. 11(a), following the extraction of the positive sequence component, the originally distorted back EMF becomes more ideally orthogonal with identical amplitude. Furthermore, Fig. 11(b) illustrates that when the switch flag is set to 1, signifying the employment of the ISOGIs, there is a notable reduction in the estimation errors pertaining to rotor speed and position, in comparison to the employment of the original back EMF estimate.

Fig. 12 shows the Lissajous waveform of the  $\alpha$ - $\beta$  frame currents with different sensorless control schemes, and the

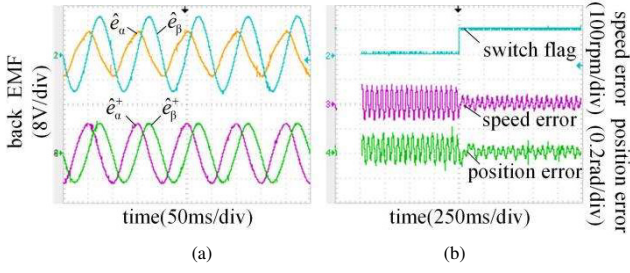


Fig. 11. Experimental results of (a) Back EMF estimates and its positive sequence component; and (b) rotor speed and position estimation errors before and after the employment of ISOGIs

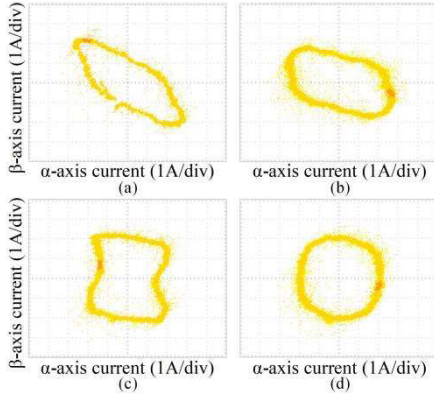


Fig. 12. Lissajous waveform of  $\alpha$ - $\beta$  frame currents (a) with no improvement; (b) with ISOGIs; (c) with open-loop neutral-voltage compensation method; (d) with the presented adaptive neutral-voltage compensation method

experimental results are recorded using the oscilloscope's XY mode. As depicted in Fig. 12(a), it is apparent that without any voltage compensation, the current distortion is pronounced, with the corresponding current trajectory significantly deviating from the desired circular pattern. In Fig. 12(b), this asymmetric condition is alleviated somewhat with the incorporation of the positive-sequence extraction. Upon comparing Figs. 12(c) and (d), it becomes evident that the employment of adaptive neutral-voltage compensation is more favorable compared to the conventional open-loop compensation. Specifically, the current phase locus approaches a more ideal circular form, and the motor imbalance is effectively mitigated. This finding is further supported by comparisons between Figs. 13(a) and (b), where in the former, the amplitude of  $\hat{e}_\alpha$  surpasses its orthogonal counterpart, suggesting an over/under compensation in the neutral voltage. On the contrary, the back EMF estimate in Fig. 13(b), which utilizes adaptive neutral-voltage compensation, exhibits a closer resemblance to a balanced state. Consequently, it is discernible that the errors in rotor speed and position estimation are further diminished.

The extraction of the positive sequence component helps reduce the rotor position estimation errors caused by asymmetric operation, while adaptive neutral-voltage compensation directly addresses the asymmetry of the faulty motor, thereby improving rotor position estimation accuracy. Combining these two methods results in superior sensorless control performance. For a more intuitive explanation, Fig. 14

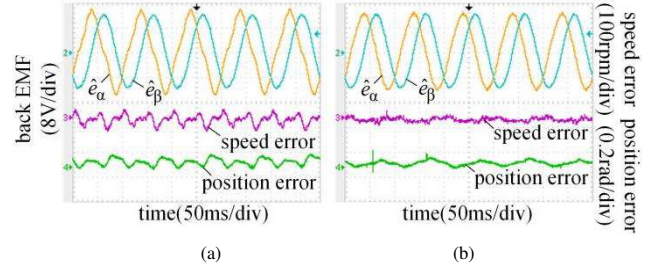


Fig. 13. Waveforms of the back EMF estimate, speed and position estimation error under (a) open-loop neutral-voltage compensation; and (b) the presented adaptive neutral-voltage compensation

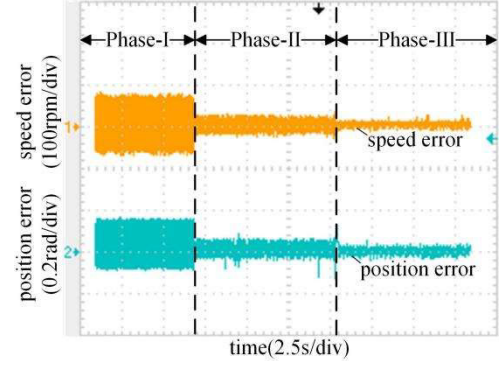


Fig. 14. Waveforms of speed and position estimation error by successively engaging three different methods

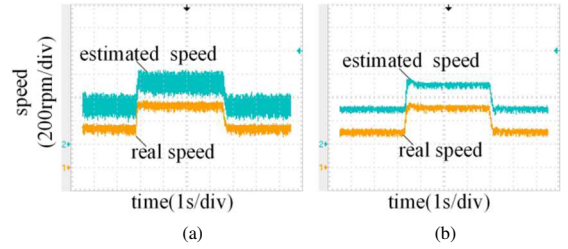


Fig. 15. Waveforms of estimated speed and real speed under (a) SMO combined with LPF; (b) SMO combined with ISOGIs

provides the profiles of the rotor speed and position estimation errors, by successively engaging three different methods for the back-EMF estimate. Phase-I stands for no improvement measures taken, Phase-II stands for the engagement of only the positive sequence extraction method, and Phase-III stands for the employment of the proposed composite method. From this figure, the speed and position estimation errors during phase-III are minimal among the three different schemes, indicating that the proposed method achieves higher position accuracy and better performance in sensorless control.

Finally, in order to verify the dynamic speed regulation performance of the motor under the improved control method, the speed regulation experiment under single-phase open-circuit fault is carried out. Fig. 15 shows the waveforms of the estimated and real speed under two types of back EMF extraction methods. The reference speed goes up from 300 rpm to 500 rpm and then down to 300 rpm. By comparing Figs. 15(a) and (b), it can be clearly seen that in the case of single-phase open circuit and variable-speed operation, the

accuracy of speed estimation can be greatly improved by using the ISOGIs. Specifically, in Fig. 15(a), the real speed fluctuates within 280~360 rpm, while the estimated speed varies across a wider 200~440 rpm band. Conversely in Fig. 15(b), the real speed fluctuates within 280~320 rpm, and the estimated speed demonstrates close tracking performance, confining its fluctuations to the identical 280~320 rpm range. In addition, the rapidity and stability of the response reflected in the waveforms in Fig. 15(b) illustrates the good dynamic performance of the proposed composite method.

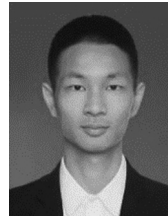
## 5. Conclusions

This paper presents a novel sensorless control method for an open-phase 5Ph-PMSM. It uses ISOGIs, instead of traditional SOGIs, to extract the positive sequence of the back EMF estimate output by the SMO, aiming to improve the accuracy of position estimation. Furthermore, considering the behavior of the neutral voltage during an open-phase fault, an adaptive neutral-voltage compensation is introduced to mitigate the asymmetry of the faulty motor, thereby significantly reducing the second and fourth harmonic of the position estimate. Experimental results validate that the proposed composite method achieves superior sensorless control performance.

## References

- (1) J. Chen, B. Tian, Y. Li, et al.: "A Sensorless Control Method for an Open-phase Five-phase PMSM by Using Series-SOGI and an Adaptive Neutral-voltage Compensation", In 27th International Conference on Electrical Machines and Systems (ICEMS2024-Fukuoka), Fukuoka, Japan (2024)
- (2) H. Zhou, W. Zhao, G. Liu, R. Cheng, and Y. Xie: "Remedial Field-Oriented Control of Five-Phase Fault-Tolerant Permanent-Magnet Motor by Using Reduced-Order Transformation Matrices", IEEE Transactions on Industrial Electronics, Vol.64, No.1, pp.169–178 (2017)
- (3) B. Tian, Q.-T. An, J.-D. Duan, D.-Y. Sun, L. Sun, and D. Semenov: "Decoupled Modeling and Nonlinear Speed Control for Five-Phase PM Motor Under Single-Phase Open Fault", IEEE Transactions on Power Electronics, Vol.32, No.7, pp.5473–5486 (2017)
- (4) L. Zhang, S. Deng, X. Zhu, and Z. Xiang: "New Flux Intensifying Technique for Five-Phase Fault-Tolerant Interior Permanent Magnet Motors Under Multiple Sensorless Operation", IEEE Transactions on Industrial Electronics, doi: 10.1109/TIE.2024.3363740.
- (5) L. Zhang, Y. Fan, C. Li, A. Nied, and M. Cheng: "Fault-Tolerant Sensorless Control of a Five-Phase FTFSCW-IPM Motor Based on a Wide-Speed Strong-Robustness Sliding Mode Observer", IEEE Transactions on Energy Conversion, Vol.33, No.1, pp.87–95 (2018)
- (6) Z. Wu, C. Cheng, W. Hua, H. Zhang, M. Hu, and H. Yin: "A Novel Square-Wave Voltage Injection Based Sensorless Control Method Robust to Single Open-Phase Fault for Dual Three-Phase IPM", IEEE Transactions on Power Electronics, Vol.38, No.6, pp.7514–7524 (2023)
- (7) J. Xu, Y. Du, B. Zhang, H. Fang, H. Guo, and Y.-H. Chen: "Sensorless Fault-Tolerant Control With Phase Delay Compensation for Aerospace FTPMSM Drives With Phase Open-Circuit and Short-Circuit Faults", IEEE Transactions on Industrial Electronics, Vol.68, No.6, pp.4576–4585 (2021)
- (8) B. Tian, M. Molinas, Q. An, B. Zhou, and J. Wei: "Freewheeling Current-Based Sensorless Field-Oriented Control of Five-Phase Permanent Magnet Synchronous Motors Under Insulated Gate Bipolar Transistor Failures of a Single Phase", IEEE Transactions on Industrial Electronics, Vol.69, No.1, pp.213–224 (2022)
- (9) G. Wang, et al.: "Enhanced Position Observer Using Second-Order Generalized Integrator for Sensorless Interior Permanent Magnet Synchronous Motor Drives", IEEE Transactions on Energy Conversion, Vol.29, No.2, pp.486–495 (2014)
- (10) G. Zhang, et al.: "Adaptive Step-Size Predictive PLL Based Rotor Position Estimation Method for Sensorless IPMSM Drives", IEEE Transactions on Power Electronics, Vol.39, No.5, pp.6136–6147 (2024)
- (11) G. Zhang, et al.: "PR Internal Mode Extended State Observer-Based Iterative Learning Control for Thrust Ripple Suppression of PMSM Drives", IEEE Transactions on Power Electronics, Vol.39, No.8, pp.10095–10105 (2024)
- (12) L. Zhang, S. Deng, X. Zhu, and Z. Xiang: "New Flux Intensifying Technique for Five-Phase Fault-Tolerant Interior Permanent Magnet Motors Under Multiple Sensorless Operation", IEEE Transactions on Industrial Electronics, doi: 10.1109/TIE.2024.3363740.
- (13) R. Zhao, Z. Xin, P.C. Loh, and F. Blaabjerg: "A Novel Flux Estimator Based on Multiple Second-Order Generalized Integrators and Frequency-Locked Loop for Induction Motor Drives", IEEE Transactions on Power Electronics, Vol.32, No.8, pp.6286–6296 (2017)
- (14) S. Maekawa, M. Sugimoto, K. Ishida, M. Nogi, and M. Kanamori: "Stability Analysis of Sensorless Speed Control for PMSM Considered Current Control System", IEEE Journal of Industry Applications, Vol.8, No.4, pp.736–744 (2019)
- (15) R.H. Loewenherz, S.A. Koschik, M. Kruse, and R.W. De Doncker: "Transformations of Reference Frames and Model Development for Multi-Phase Machines", IEEE Journal of Industry Applications, Advance online publication (2021)
- (16) S. Maekawa, M. Wu, and H. Kubota: "Position Sensorless Control Method for Permanent Magnet Synchronous Motors Using Speed Observer and Opened Phase Voltage Caused by Magnetic Saliency", IEEE Journal of Industry Applications, Vol.8, No.6, pp.875–883 (2019)

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